

Method Selection Guide

PCA →

- ⑩ **It does** → PCA finds linear direction that explain the largest amount of variance.
- ⑩ **Biggest Strength** → PCA fast and reversible using ``inverse_transform``.
- ⑩ **Biggest Weakness** → PCA is linear, so non linear structures may not be separated clearly.

t-SNE →

- ⑩ **It does** → It creates a low-dimensional representation that focuses strongly on preserving local neighbourhood relationships.
- ⑩ **Biggest Strength** → It can produce very clear visual separation of local groups.
- ⑩ **Biggest Weakness** → Global distances and apparent cluster sizes should not be interpreted as literal properties of the original data.

UMAP →

- ⑩ **It does** → It creates a non-linear low-dimensional representation that focuses on local structure while often retaining useful larger-scale structure.
- ⑩ **Biggest Strength** → It provides useful visualizations and is often practical for larger datasets.
- ⑩ **Biggest Weakness** → The result depends on parameters and randomness, and the 2D geometry should not automatically be treated as the original geometry.

Method Selection

<u>Sr. NO.</u>	<u>Situation</u>	<u>Recommendation Method</u>	<u>Reason</u>
1.	Preprocessing before a model	PCA	Compact numerical representation and dimensionally reduction
2.	Exploring an unfamiliar dataset	UMAP/t-SNE	Good for discovering local groups
3.	Compressing data for storage	PCA	Supports reconstructio
4.	Making a presentation figure	UMAP/t-SNE	Often gives clearer visual separation

⑩ Two claims that must never be made from a t-SNE plot

- ↘ **Claim 1 :-** The distance between two clusters represents their true distance in the original feature space.
 - ✦ This is not reliable because t-SNE does not preserve global distances.
- ↘ **Claim 2 :-** This cluster is larger or more variable because it occupies more spaces in the t-SNE plot.
 - ✦ This is not reliable because t-SNE can change apparent cluster size and density while preserving local relationships.

UMAP warm-up trap :- UMAP may compile the code during its first execution. Therefore the first execution can look much slower than later execution.

Run a small warm-up first, discard that timing, and then measure the full-data run.

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Final Rule :-

- ⑩ PCA is useful when variance preservation, preprocessing, compression or reconstruction matters.
- ⑩ t-SNE and UMAP are useful mainly for exploring and visualizing local structure.
- ⑩ A beautiful embedding is evidence about grouping and neighbourhoods, not automatic evidence about original-space distance, cluster size, or variance.